Documentation for Android Native SDK

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introduce

This document describes how to use the iHealth Device SDK to accomplish the major operation:

Authentication
Scan Device
Connect Device
Online Measurement
Offline Measurement
iHealth Device Management

Latest version

2.15.1

Update Notes

Version 20250924 （2025/09/24）

1. Update libiHealth.so to adapt to 16KB Page

Note: Only the .so files in jniLibs need to be kept, and the others like libECGOffline.so、libECGOnline.so、libVeryFitMulti.so need to be removed.

Version 20250813 (2025/08/13)

2. Adapt to 16KB Page

3. Fixed other bugs

Version 20250610 (2025/06/10)

1. Adapt to Android 15

2. Fixed other bugs

Version 20241029 (2024/10/29)

1. Support BP5S hardware 2.0.0 API changes in detail please refer to https://chenxuewei-ihealth.github.io/ihealthlabs-sdk-docs/docs/android/blood_pressure/bp5s.

2. Fixed other bugs

Version 20240425 (2024/04/25)

1. Support KD-723 firmware 1.1.0 API changes in detail please refer to https://chenxuewei-ihealth.github.io/ihealthlabs-sdk-docs/docs/android/blood_pressure/kd723.

2. Fixed other bugs

Version 20240102 (2024/01/02)

1. Support HS2SPro

2. Fixed other bugs

Version 20230911 (2023/09/11)

1. Support new device AM6.

2. Support KN550-BT firmware 2.0.1. API changes in detail please refer to https://chenxuewei-ihealth.github.io/ihealthlabs-sdk-docs/docs/android/blood_pressure/kn550.

3. Fixed BG5S and KN550-BT issues directly without scanning. BG5S and KN550-BT(2.0.0) use the same UUID. If the direct connection cannot determine the device type without scanning, the parameter passed by the caller is read to determine the type.

4. Delete BP3M related code.

5. Support PO3 latest firmware 3.1.1. Limit PO3 PI range from 0.2 to 20 API change in detail please refer to https://chenxuewei-ihealth.github.io/ihealthlabs-sdk-docs/docs/android/blood_oxygen/po3.

Version 2022121200 (2022/12/12)

1. Optimize the encryption scheme of HS6
2. Fix some other known problems

Version 2021110300 (2021/11/03)

1. Support new KN-550BT.

Version 2021102200 (2021/10/22)

1. Fix BG5S connection issue for special version UUID.
2. Fix Hs2s result don't have skeletal muscle for above hardware version 6.0.0.

Version 2021091500 (2021/09/15)

1. Fix BG5S connection issue

Version 2021080900 (2021/08/09)

1. Fix Hs2s Body moisture issue

Version 2021071200 (2021/07/12)

1. Support HS2SPro
2. Fix BG5S issues

Version 2021020900 (2021/02/09)

1. Support BG5S

Version 2020121100 (2020/12/11)

1. Remove the trust manager function

Version 2020111100 (2020/11/11)

1. Hs2s support heart rate measurement

Version 2020100900 (2020/10/09)

1. SDK upgrade to version 2.9.0
2. Add new device PT3 SBT.
3. Add new device PO1.
4. Upgrade Android API29
5. Support Androidx Library.

Version 2020091800 (2020/09/18)

1. SDK upgrade to version 2.8.5
2. Fix NT13B issue.

Version 2020041400 (2020/04/14)

1. SDK upgrade to version 2.8.0
2. Fix BG1, BG5, BG5S issue.

Version 2019121300 (2019/12/13)

1. SDK upgrade to Version 2.7.1
2. support NT13B equipment
3. Optimize device connectivity need to refer to four

Version 2019081600 （2019/8/16）

1. SDK upgrade to Version 2.6.0
2. Change initialization method: initialize in application
3. Now we just need to introduce a jar into lib to work properly, whereas we used to need to refer to four
4. Support HS2S
5. Support AM5

Version 2019032000 （2019/3/20）

1. Release Description Document
2. SDK upgrade to Version 2.5.1

Support iHealth Device for Android

AM:

iHealth AM3

iHealth AM3S(Edge)

iHealth AM4(Wave)

BG:

iHealth BG1(Align)

iHealth BG5(Smart)

iHealth BG5S(Gluco+)

BP:

iHealth BP3L(Ease)

iHealth BP3M

iHealth BP5(Feel)

iHealth BP7

iHealth BP7S(View)
iHealth KN-550BT(Track)
iHealth BPM1
iHealth ABP100
iHealth ABI
iHealth BP5S

Continua BP:
iHealth KD723
iHealth KD926

HS:
iHealth HS3
iHealth HS4(Lite)
iHealth HS4S
iHealth HS5
iHealth HS6(Core)
iHealth HS2(Lina)
iHealth HS2S

PO:
iHealth PO3/PO3M(Air)
PO1

Thermometer:
THV3(FDIR-V3)
TS28B
NT13B
PT3SBT

Support phone os

Support phone os 4.0+:
ABI, BP3M, BP5, BP7, BP7S, BPM1, BG1, BG5, HS3, HS4S, HS5, HS6

Support phone os 4.3+:
AM3, AM3S, AM4, AM5, BP3L, BP550BT, KD723, KD926, ABP100, HS4, PO1, PO3,
HS2,BG5S, BP5S, ECG3, THV3, TS28B, HS2S,NT13B, PT3SBT

connection mode

iHealth Device SDK communicate with iHealth Device
by USB, Bluetooth, BluetoothLe, Wifi or Audio.

USB:
iHealth BP3M iHealth ECGUSB

Bluetooth:
iHealth BP5, iHealth BP7, iHealth BP7S,
iHealth ABI, iHealth HS3, iHealth HS4S, iHealth BG5,

BluetoothLe:
iHealth AM3, iHealth AM3S, iHealth AM4, iHealth AM5,
iHealth BP3L, iHealth KN-550BT, iHealth KD723, iHealth KD926,
iHealth ABP100, iHealth HS4, iHealth PO1, iHealth PO3(PO3M),
iHealth HS2, THV3,TS28B,iHealth BG5S,iHealth ECG3,iHealth BP5S
iHealth HS2S,NT13B, PT3SBT

Audio:
iHealth BG1

Wifi:
iHealth BPM1, iHealth HS5, iHealth HS6

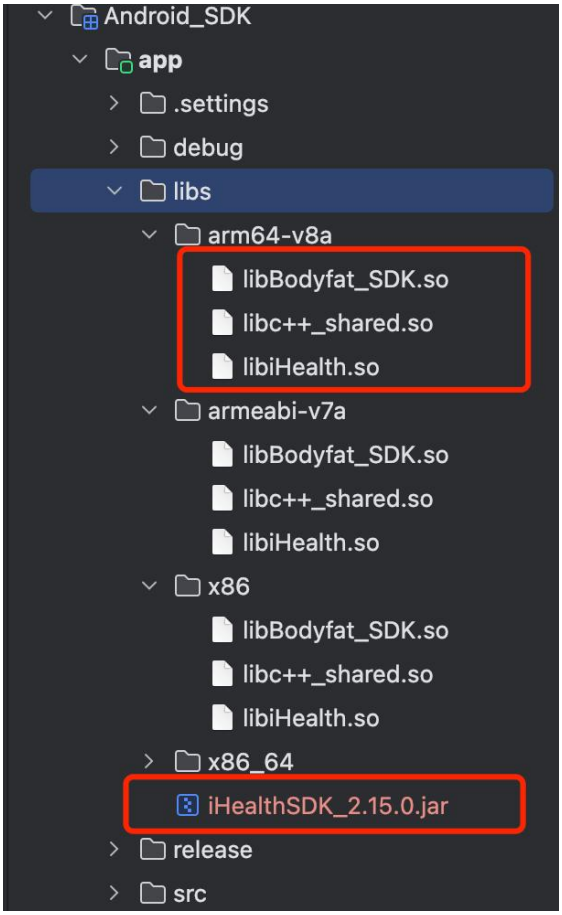
How to import the ihealth SDK

Registration application
First of all, we need to register your application on the website. After waiting for the approval, we can download the. PEM files necessary for authentication.

For more details, please click the link below. Getting start
SDK download and import When you create an application that passes the audit， you can download the NativeSDK. The downloaded file contains the Demo and the .so files and .jar files that required. Put them in the Android project libs directory

Download

Download SDK
libs: Add the following code



build.gradle: Add the following code

```
buildscript {

    repositories {
        google()
        mavenCentral()
        maven { url 'https://jitpack.io' }
    }
    dependencies {
        classpath 'com.android.tools.build:gradle:8.9.1'
        classpath 'org.greenrobot:greendao-gradle-plugin:3.2.2'
//        classpath 'com.jakewharton:butterknife-gradle-plugin:10.2.3'
    }
}

allprojects {
    repositories {
        google()
        mavenCentral()
        maven { url 'https://jitpack.io' }
    }
}
```

classpath 'org.greenrobot:greendao-gradle-plugin:3.2.2'

```
apply plugin: 'com.android.application'

android {
    compileSdkVersion 28
    defaultConfig {
        applicationId "com.demo.sdk"
        minSdkVersion 18
        targetSdkVersion 28
        versionCode 1
        versionName "1.0.0"
        testInstrumentationRunner "android.support.test.runner.AndroidJUnitRunner"

        multiDexEnabled true
        ndk {
            abiFilters 'armeabi', 'armeabi-v7a', 'arm64-v8a', 'x86', 'x86_64'
        }
    }
    buildTypes {
        release {
            minifyEnabled false
            proguardFiles getDefaultProguardFile('proguard-android.txt'), 'proguard-rules.pro'
        }
    }
    sourceSets {
        main {
            jniLibs.srcDirs = ['libs']
        }
    }
}

dependencies {
    api fileTree(include: ['*.jar'], dir: 'libs')
    api 'com.android.support:appcompat-v7:28.0.0'
    api 'com.android.support.constraint:constraint-layout:1.1.3'
    api 'com.android.support:support-v4:28.0.0'
    api 'com.android.support:design:28.0.0'

    testImplementation 'junit:junit:4.12'
    androidTestImplementation 'com.android.support.test:runner:1.0.2'
    androidTestImplementation 'com.android.support.test.espresso:espresso-core:3.0.2'
    api project(path: ':baselibrary')
    //黄油刀
    api 'com.jakewharton:butterknife:8.8.1'
    annotationProcessor 'com.jakewharton:butterknife-compiler:8.8.1'
    implementation files('libs/iHealthSDK_2.6.0.jar')

    implementation 'com.alibaba:fastjson:1.2.48'
    implementation 'com.squareup.okhttp3:okhttp:3.12.1'
    implementation 'com.squareup.okio:okio:1.15.0'
    implementation 'com.google.code.gson:gson:2.8.5'
    implementation 'org.greenrobot:greendao:3.2.2'
    implementation 'no.nordicsemi.android:dfu:1.6.1'
    implementation 'com.android.support:multidex:1.0.3'

}
```

multiDexEnabled true

ndk {

abiFilters 'armeabi', 'armeabi-v7a', 'arm64-v8a', 'x86', 'x86_64'

}

sourceSets {

main {

jniLibs.srcDirs = ['libs']

}

}

implementation 'com.alibaba:fastjson:1.2.48'

implementation 'com.squareup.okhttp3:okhttp:3.12.1'

implementation 'com.squareup.okio:okio:1.15.0'

implementation 'com.google.code.gson:gson:2.8.5'

implementation 'org.greenrobot:greendao:3.2.2'

implementation 'no.nordicsemi.android:dfu:1.6.1'

implementation 'com.android.support:multidex:1.0.3'

Manifest: Add the following code

Add the following code：

```
<service
    android:name="com.ido.ble.bluetooth.DeviceConnectService"
    android:exported="false" />
```

```
<service
    android:name="com.ido.ble.dfu.DFUService"
    android:enabled="true"
    android:exported="false"/>
```


```
<service
    android:name="com.ido.ble.bluetooth.DeviceConnectService"
    android:exported="false" />

<service
    android:name="com.ido.ble.dfu.DFUService"
    android:enabled="true"
    android:exported="false" />
```


How to use the iHealth SDK

Configure

Specific configuration as shown below:
AndroidManifest.xml Permissions:

```
<!-- Internet communication and detect / manage Wi-Fi state -->
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE" />
<uses-permission android:name="android.permission.CHANGE_WIFI_MULTICAST_STATE"
/>

<!-- Bluetooth permission for communicating with iHealth devices -->
<uses-permission android:name="android.permission.BLUETOOTH" />
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />

<!-- Permission for writing on device memory -->
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />

<!-- Permission for locating user device in order to search beacons -->
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />

<!-- Permission for iHealth BG1 device (which connect with phone by headphone jack) -->
<uses-permission android:name="android.permission.RECORD_AUDIO"
<uses-permission android:name="android.permission.MODIFY_AUDIO_SETTINGS" />
```

Note: Android 6.0 + OS needs to apply for dynamic permissions at the same time Applying for location permissions is necessary for Android operating systems above 6.0 to apply for Bluetooth permissions.
Mobile memory read and write permission is for users to generate logs and send them to us in order to solve problems.

Example：

Operation procedure for BP5.
1. Initialization iHealth SDK.
We recommend initialization in application

```
/*
* Initializes the iHealth devices manager. Can discovery available iHealth devices nearby
* and connect these devices through iHealthDevicesManager.
*/
iHealthDevicesManager.getInstance().init(application, Log.VERBOSE, Log.WARN);
```

Note: if you replace
ihealthDevicesManager.init (application, Log.VERBOSE, Log.VERBOSE)
at initialization. You can get a debug file that name is ihealthSDK-XX_SDK_Debug.txt at path Android -data -package name -files -ihealthSDK-XX_SDK_Debug.txt If you encounter problems that you can't solve, please send us this file.
2. Register callback, and get a callback ID.

```
/*
* Register callback to the manager. This method will return a callback Id.
*/
int callbackId =
iHealthDevicesManager.getInstance().registerClientCallback(iHealthDevicesCallback;
```

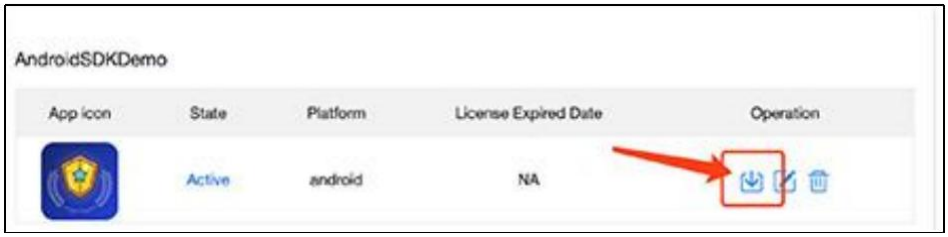
3. Add callback filter.
For example, if you only care about one or more of your devices, you can set these filters

```
iHealthDevicesManager.getInstance().addCallbackFilterForAddress(clientCallbackId, ...);
iHealthDevicesManager.getInstance().addCallbackFilterForDeviceType(clientCallbackId, ...);
```

4. Authentication.
If you want to use the iHealth Device, you must first call authentication method before connect a device.

```
iHealthDevicesManager.getInstance().sdkAuthWithLicense(byte[] licenseDataBuffer);
```

Where to get the licenseDataBuffer?
1)download the .pem file form the website after register your app. You can download .pem file at here



2)integrate the pem file into your project.(Under the Assets) 3)read the binary data. 4)if you do not have this button,Please wait for the audit patiently。
Example:

```
        try {
                InputStream is = getAssets().open("license.pem");
                int size = is.available();
                byte[] buffer = new byte[size];
                is.read(buffer);
                is.close();
                boolean isPass = iHealthDevicesManager.getInstance().sdkAuthWithLicense(buffer);
                Log.i("info", "isPass:
                " + isPass);
        } catch (IOException e) {
                e.printStackTrace();
        }
```

If authenticate success(isPass), all the api avaliable,
else please check you .pem file on website.
5. Discovery an iHealth device or multi devices.

```
iHealthDevicesManager.getInstance().startDiscovery(DiscoveryTypeEnum type);
static
{
```

```
        DiscoveryTypeEnum[] typeArray = DiscoveryTypeEnum.values();
        DEVICE_DISCOVERY_TYPE_ARRAY = new String[typeArray.length];
        for (int i = 0; i < typeArray.length; i++) {
        DEVICE_DISCOVERY_TYPE_ARRAY[i] = typeArray[i].name();
        }
}
...
        String name = DEVICE_DISCOVERY_TYPE_ARRAY[position];
        for (DiscoveryTypeEnum type : DiscoveryTypeEnum.values()) {
                if (name.equals(type.name())) {
                iHealthDevicesManager.getInstance().startDiscovery(type);
                return;
                }
        }
```

Register callback methods to monitor search status. If you search for a device, it return the device MAC address device type

```
private iHealthDevicesCallback iHealthDevicesCallback = new iHealthDevicesCallback() {
        @Override
        public void onScanDevice(String mac, String deviceType) {
        }
};
```

6. Connection an iHealth device.

```
iHealthDevicesManager.getInstance().connectDevice(userName, mac, type);
private iHealthDevicesCallback iHealthDevicesCallback = new iHealthDevicesCallback() {
        @Override
        public void onDeviceConnectionStateChange(String mac, String deviceType, int
        status) {
                if (status == iHealthDevicesManager.DEVICE_STATE_CONNECTED) {
                        // msg.what = HANDLER_CONNECTED;
                        // connect device successful
                } else if (status == iHealthDevicesManager.DEVICE_STATE_DISCONNECTED) {
                        // msg.what = HANDLER_DISCONNECT;
                        // device disconnect
                } else if (status == iHealthDevicesManager.DEVICE_STATE_CONNECTIONFAIL)
                {
                // connect device fail
                } else if (status == iHealthDevicesManager.DEVICE_STATE_RECONNECTING) {
                        // Automatically retry the connect device
                }
                 // msg.setData(bundle);
        }
};
```

7. Get iHealth device controller.

When iHealthDevicesCallback return to DEVICE_STATE_CONNECTED, you can continue to create Control to continue operating the device.

```
/*
* Get Bp5 device controller
*/
Bp5Control bp5Control = iHealthDevicesManager.getInstance().getBp5Control(mac);
```

8. Invoke some api function.

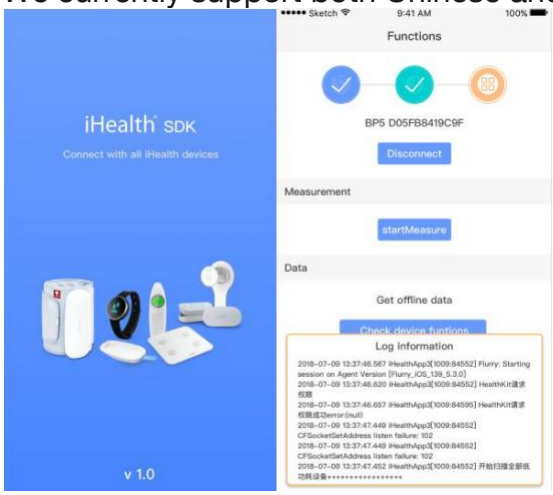
```
/*
* Get the Bp5 device battery level
*/
bp5Control.getBattery();
```

After BP5 response, your can get the battery level value with callback onDeviceNotify and specific action {@link BpProfile#ACTION_BATTERY_BP}

```
private iHealthDevicesCallback iHealthDevicesCallback = new iHealthDevicesCallback() {
        @Override
        public void onDeviceNotify(String mac, String deviceType, String action, String message) {
        }
};
```

New Demo:

We created a new demo project to help developers better access SDK.
We currently support both Chinese and English.



New Demo Latest version

Version 1.2.9

New Demo Support iHealth Device for Android

BP:
iHealth BP5(Feel)
iHealth KN-550BT(Track)
iHealth BP7S(View)
iHealth BP3L(Ease)

BG:
iHealth BG1(Align)
iHealth BG5(Smart)
iHealth BG5S(Gluco+)

HS:
iHealth HS2(Lina)
iHealth HS4(Lite)
iHealth HS6(Core)
iHealth HS2S

AM:
iHealth AM3S(Edge)
iHealth AM4(Wave)
iHealth AM5

Thermometer:
TS28B
THV3(FDIR-V3)
PT3SBT

PO:
iHealth PO1
iHealth PO3/PO3M(Air)

FAQ for Android

1.Android permission explaination

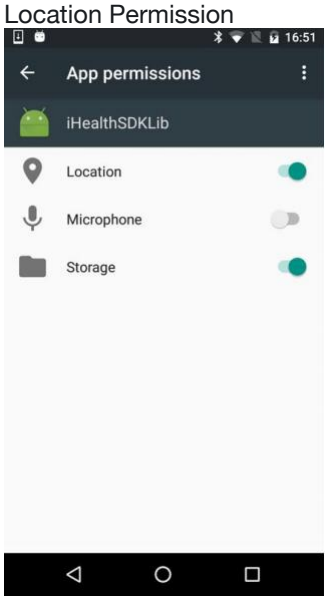
```
<!-- Internet communication and detect / manage Wi-Fi state -->
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE" />
<uses-permission android:name="android.permission.CHANGE_WIFI_MULTICAST_STATE"
/>
<!-- Bluetooth permission for communicating with iHealth devices -->
<uses-permission android:name="android.permission.BLUETOOTH" />
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />
<!-- Permission for writing on device memory -->
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
<!-- Permission for locating user device in order to search beacons -->
<!--
https://developer.android.com/reference/android/bluetooth/le/BluetoothLeScanner.html#start
tScan(android.bluetooth.le.ScanCallback) -->
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<!-- Permission for iHealth BG1 device (which connect with phone by headphone jack) -->
<uses-permission android:name="android.permission.RECORD_AUDIO"
<uses-permission android:name="android.permission.MODIFY_AUDIO_SETTINGS" />
```

2.Android 6.0 location permission

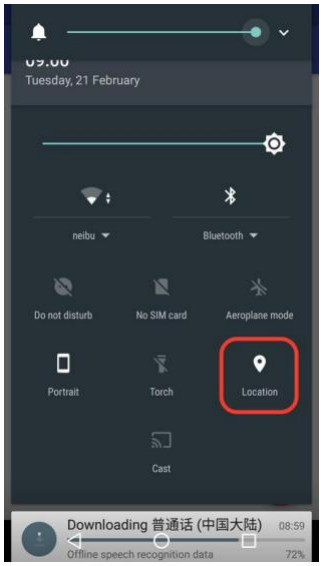
If targetSdkVersion >= 23, need location permission as follows:

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
```

Turn on location function for the phone.
Location Permission



Location Function



3.Android 7.0 connect problem

Connect will fail on Android 7.0 with devices as follows:
1. HS4 Firmware Version < 1.0.5
2. PO3 (Firmware Version < 1.0.5 || (Firmware Version = 1.0.5 && Display Version != 11.0.5))
3. PO3M Firmware Version < 2.1.4
4. BP3L (Firmware Version =1.0.0 && BT Version < 1.2.4)
5. KN-550BT (Firmware Version =1.0.0 && BT Version < 1.2.4)
6. AM3 All the version
7. AM3S Firmware Version < 1.1.8
8. AM4 Firmware Version < 1.4.4

4.Android 8.0 connect problem

Connect will fail on Android 8.0 with devices as follows:
1. HS4 BT Version <= 1.0.8
2. PO3 BT Version <= 1.0.8
3. PO3M BT Version <= 1.0.8
4. BP3L BT Version <= 1.2.4
5. KN-550 BT Version <= 1.2.4
```
```